Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination: A Diagnostic Analysis

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Abstract

Temporal abstraction is a long-sought capability in reinforcement learning, and world-model architectures offer a natural testbed: one can modify the temporal resolution of imagined rollouts without changing environment interaction. We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, revealing a failure mode we term *Conditioning-Induced Value Overestimation* (CIVO). Through controlled factorial experiments on Walker Walk and Cheetah Run (DeepMind Control), we identify the *trained* Δt-embedding as the necessary and sufficient architectural component for CIVO: when the critic receives a learned temporal-step embedding, the Normalized Value Excess (NVE) reliably exceeds 1.5× in Walker Walk (2/2 seeds) and triggers with high probability in Cheetah Run (1/3 seeds exceed 1.5×; all 3 seeds show degradation averaging 34%). Removing or freezing the Δt-embedding eliminates CIVO across all tested seeds (0/11 exceed 1.5×), confirming the causal role. An RSSM fidelity analysis shows that the mechanism scales with environment dynamics complexity: Walker Walk (obs gap 2.04×) exhibits stronger CIVO than Cheetah Run (obs gap 1.80×). Restricting imagination to Δt=1 eliminates the classic CIVO pathway but does not prevent collapse entirely—2/4 seeds still collapse via a distinct, eval-driven pathway with rising actor entropy. Our findings contrast with TAWM, where Δt-conditioning succeeds in MPC-based planning, suggesting the vulnerability is specific to imagination-based actor-critic training. We propose the NVE metric (AUROC $\geq$ 0.93 across environments) as a practical diagnostic for detecting CIVO in world-model architectures.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a foundational goal in reinforcement learning [Sutton et al., 1999, Precup, 2000]. World-model architectures such as DreamerV3 [Hafner et al., 2023] offer a compelling substrate for temporal abstraction: by modifying the temporal resolution Δt of imagined rollouts within a learned latent-dynamics model, one can explore multi-step planning without altering the agent's environment interaction protocol.

Despite this appeal, our systematic investigation reveals that *all* variable-rate imagination configurations in DreamerV3 degrade performance relative to the Δt=1 baseline. Fixed Δt=2 imagination peaks at 485 eval return before collapsing to ∼135 (vs. baseline ∼950); fixed Δt=4 never exceeds 177; a learned gating mechanism peaks at 541 before collapsing to 99. The question is not *whether* variable-rate imagination fails, but *why*.

We trace the root cause to a specific architectural component: the **trained Δt-embedding**. When DreamerV3's critic receives a learned embedding that encodes the temporal step size, it develops *Conditioning-Induced Value Overestimation* (CIVO)—a positive feedback loop where the critic exploits the Δt-conditional information to systematically overestimate values for Δt>1 imagined states. We introduce the *Normalized Value Excess* (NVE) metric, defined as the ratio of an experiment's value-to-eval ratio to the baseline's value-to-eval ratio at the same training step, and show that NVE > 1.5× reliably diagnoses CIVO onset.

Our controlled factorial experiments across Walker Walk and Cheetah Run yield three core findings:

1. **Trained Δt-embedding is necessary and sufficient for CIVO.** All seeds with trained Δt-embedding show NVE > 1.0×; the 1.5× threshold achieves 100% accuracy in Walker Walk (4/4) and 80% accuracy cross-environment (8/10). Removing or freezing the Δt-embedding eliminates CIVO across 11/11 seeds.
2. **CIVO severity scales with environment dynamics complexity.** The RSSM's observation prediction gap between Δt=1 and Δt=2 is 2.04× in Walker Walk vs. 1.80× in Cheetah Run; Walker exhibits stronger CIVO (mean peak NVE 2.07×) than Cheetah (1.74×).

3. Δt=1 **imagination eliminates CIVO but not all failure modes.** Restricting imagination to Δt=1 while retaining the trained Δt-embedding prevents the value-driven CIVO pathway, but 2/4 seeds still collapse via a distinct eval-driven pathway with rising actor entropy.

Our findings stand in instructive contrast to TAWM [Nhu et al., 2025], where Δt-conditioning succeeds in MPC-based planning (TD-MPC2). The key difference is that MPC re-plans from real states at each environment step, limiting the compounding of model errors, while DreamerV3's imagination-based actor-critic trains entirely within the learned model, allowing CIVO to self-reinforce over extended rollouts. This architecture-specific vulnerability—invisible to MPC-based methods—motivates caution when transferring temporal-abstraction techniques across planning paradigms.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] and its extensions formalize temporal abstraction as policies over extended time scales. FiGAR [Sharma et al., 2017] learns action repetition counts; Skip-DQN [Dabney et al., 2021] treats the skip interval as part of the action. These model-free approaches sidestep the prediction-quality issues we identify, since they do not rely on a learned dynamics model for multi-step imagination.

World-model RL. The Dreamer family [Hafner et al., 2020, 2021, 2023] learns a Recurrent State-Space Model (RSSM) and trains actor and critic entirely within imagined rollouts. IRIS [Micheli et al., 2023] and TWM [Robine et al., 2023] explore transformer-based alternatives. All share the property that imagination quality bounds policy quality—a property central to our analysis.

Temporal abstraction in world models. TAWM [Nhu et al., 2025] introduces Δt-conditioning in TD-MPC2's MPC framework and reports successful variable-rate planning. Critically, TD-MPC2 re-plans from real states, bounding model-error compounding. Our work shows that the same Δt-conditioning mechanism fails in imagination-based actor-critic training, where the critic trains entirely on model-generated data.

Value overestimation in MBRL. Model-based methods are susceptible to value overestimation from model exploitation [Janner et al., 2019]. Delusional bias [Lu et al., 2018] arises from policy-class constraints creating unrealizable Q-values. Our CIVO mechanism is distinct: it arises from *learned conditioning* (the Δt-embedding) amplifying model prediction errors into systematic value overestimation, not from policy-class constraints.

3 Methodology

3.1 DreamerV3 with Variable-Rate Imagination

We build on the standard DreamerV3 architecture. The RSSM models latent dynamics as $s_{t+1} = f_\theta(s_t, a_t)$. For variable-rate imagination, we introduce a temporal step size $\Delta t \in \{1, 2, \ldots\}$ and a Δt-embedding $e(\Delta t) \in \mathbb{R}^d$ (default d=8) that conditions the RSSM, reward head, and critic on the temporal resolution. Temporal subsampling constructs training transitions at Δt-step intervals from the replay buffer, so a Δt=2 transition maps $(s_t, a_t) \to (s_{t+2}, r_{t:t+2})$.

3.2 Factorial Experiment Design

We isolate the contribution of each architectural component through a controlled factorial design with four Δt-embedding configurations:

- **Trained** (Δt-emb trained, subsample ON): The Δt-embedding receives gradient updates from training on Δt>1 transitions. This is the CIVO-susceptible configuration.
- **Untrained** (Δt-emb present but subsample OFF): The embedding exists but is never exposed to Δt>1 data during training. Imagination still uses Δt=2.

- **Absent** (Δt-embed-dim=0, subsample ON): No Δt-embedding; the model receives $\Delta t>1$ training data but cannot condition on Δt.
- **Frozen** (Δt-emb frozen at random init, subsample ON): The embedding exists but `requires_grad=False`; the model cannot learn Δt-conditional representations.

Additionally, we test Δt=1 imagination (restricting the imagination step to Δt=1 while training data includes $\Delta t>1$ transitions) to probe whether the vulnerability lies in imagination or in the learned representations.

3.3 Normalized Value Excess (NVE)

We define the NVE as a diagnostic metric for value overestimation:

$$\text{NVE}(t) = \frac{V_{\text{exp}}(t)/R_{\text{exp}}(t)}{V_{\text{base}}(t)/R_{\text{base}}(t)} \tag{1}$$

where $V(t)$ is the mean predicted value and $R(t)$ is the mean evaluation return at training step t. NVE = 1.0 indicates calibration equivalent to the baseline; NVE > 1.5 indicates pathological overestimation. We restrict NVE computation to $t \geq 35$K steps, where the baseline value-to-eval ratio has stabilized (> 0.24), avoiding spurious amplification from small denominators at early training.

3.4 Environments

Experiments are conducted on three DeepMind Control Suite tasks: **Walker Walk** (primary, 24D observation, balancing + locomotion), **Cheetah Run** (cross-environment replication, higher dimensionality, periodic gait dynamics), and **Cartpole Swingup** (preliminary, simpler 5D observation, smooth pendulum dynamics).

4 Experiments

4.1 Walker Walk: Isolating the CIVO Mechanism

Trained Δt-embedding triggers pathological NVE. Table 1 presents the trajectory comparison for two independent seeds of the trained Δt-embedding configuration (008d). Both seeds exceed NVE > 1.5× (peak NVE: 4.12× seed 1, 2.05× seed 2), but their outcomes diverge: seed 1 collapses (eval 131 at 105K, NVE sustained > 1.5× across three consecutive checkpoints), while seed 2 self-corrects (eval 268 at 105K, NVE 0.38×). The bifurcation depends on early training stochasticity: when NVE exceeds 1.5× and *persists* across consecutive checkpoints, the positive feedback loop locks in irreversibly.

Table 1: Walker Walk trained Δt-embedding (008d): seed 1 vs. seed 2 trajectory comparison. Seed 1 sustains NVE > 1.5× and collapses; seed 2's overestimation is transient and self-corrects.

Step	S1 Eval	S1 NVE	S2 Eval	S2 NVE
25K	167	1.78×	141	2.05×
35K	146	1.71×	164	1.24×
45K	141	2.09×	141	1.62×
55K	83	4.12×	172	1.41×
65K	250	1.23×	209	0.94×
85K	202	1.15×	236	0.51×
105K	131	1.57×	268	0.38×

Causal control: removing the Δt-embedding. Table 2 shows NVE trajectories for three seeds with Δt-embed-dim=0. No seed exceeds NVE > 1.5× (peak range: 0.95–1.26×), and no collapse occurs. This confirms that the Δt-embedding is *necessary* for CIVO: without it, the critic cannot exploit Δt-conditional

information, and the overestimation feedback loop is architecturally blocked. The cost is reward prediction degradation—reward MAE increases ∼75× vs. baseline because the reward head cannot distinguish Δt=1 from Δt=2 transitions (reward gap 1.01×).

Table 2: Walker Walk Δt-embed-dim=0 (008f): NVE across 3 seeds. No seed exceeds 1.5×.

Step	S0 NVE	S1 NVE	S2 NVE
25K	1.02×	0.95×	1.26×
45K	0.83×	0.75×	0.72×
65K	0.37×	0.45×	0.53×
85K	0.65×	0.55×	0.60×
105K	0.30×	0.40×	1.14×

Untrained Δt-embedding: value growth suppression. Table 3 summarizes the untrained Δt-embedding configuration (008a). When the embedding is present but never exposed to Δt>1 training data (subsample OFF), NVE stays consistently below 1.0× across all three seeds (range: 0.55–0.90×), and the characteristic zigzag recovery pattern is reproducible. Performance reaches 74–85% of baseline (mean eval 726 at 105K vs. baseline ∼916), confirming that the untrained Δt-embedding structurally suppresses value growth without triggering CIVO.

Table 3: Walker Walk untrained Δt-embedding (008a): 3-seed summary. All seeds show structural value suppression (NVE < 1.0×) and zigzag recovery.

Seed	Eval @ 105K	NVE @ 105K	vs. Baseline	Zigzag Cycles
0	778	0.59×	−15%	2
1	720	0.62×	−21%	2
2	680	0.70×	−26%	1.5

Frozen Δt-embedding: gradient starvation. A frozen random Δt-embedding (requires_grad=False) prevents CIVO (NVE range: 0.39–0.47×, no collapse) but introduces severe gradient starvation: actor loss approaches zero, eval reaches only 204 at 105K (22% of baseline). The frozen embedding provides constant noise that the RSSM must route around, wasting representational capacity. This result completes the 2×2 ablation: {trained: CIVO risk} × {frozen: gradient starvation} × {absent: reward degradation} × {untrained: value suppression}.

Combined Walker Walk configuration comparison. Table 4 summarizes all four Δt-embedding configurations, establishing that *only* training the Δt-embedding on Δt>1 data creates CIVO.

Table 4: Walker Walk Δt-embedding ablation summary. Only the trained configuration triggers NVE > 1.5×.

Configuration	Seeds	NVE > 1.5×	Collapse Rate	Eval @ 105K
Trained	2	2/2 (100%)	1/2 (50%)	131–268
Untrained	3	0/3 (0%)	0/3 (0%)	680–778
Absent	3	0/3 (0%)	0/3 (0%)	148–503
Frozen	1	0/1 (0%)	0/1 (0%)	204
Baseline (Δt=1)	—	—	0%	∼916

4.2 Cross-Environment Replication: Cheetah Run

Baseline. Three seeds of the Δt=1 baseline converge to IQM eval 718, with value-to-eval ratio stabilizing at ∼65–70% from 135K steps onward (Table 5).

Table 5: Cheetah Run baseline (3 seeds).

Seed	Eval Peak	Eval Final (last-10 mean)	v/e Stable
0	755	738	67%
1	733	720	65%
2	725	695	66%

Trained Δt-embedding: CIVO replicates with reduced severity. Table 6 shows per-seed results. Seed 0 collapses (NVE peak 2.38× at 55K, eval nadir 270, partial recovery to 345), while seeds 1 and 2 show moderate degradation with borderline NVE (1.38× and 1.47×, respectively—below the 1.5× threshold but above 1.0×). Mean degradation is 34% vs. baseline. The reduced CIVO severity relative to Walker Walk is mechanistically expected: Cheetah Run's more periodic gait dynamics produce a smaller RSSM fidelity gap (Section 4.4), yielding weaker Δt-conditional value overestimation.

Table 6: Cheetah Run trained Δt-embedding: per-seed NVE peak and outcome.

Seed	NVE Peak	Peak Step	Eval @ 105K	vs. Baseline	Exceeds 1.5×
0	2.38×	55K	345	−52%	✓
1	1.38×	55K	568	−21%	
2	1.47×	45K	510	−29%	

Causal control: Δt-embed-dim=0. Removing the Δt-embedding in Cheetah Run replicates the Walker Walk finding cleanly: 0/3 seeds exceed NVE > 1.5× (peak range 0.72–0.85×), confirming that the causal role of the Δt-embedding generalizes across environments. Mean eval 507 at 105K (−29% vs. baseline) reflects the same reward-prediction trade-off as Walker Walk.

Untrained Δt-embedding: value suppression replicates. Three seeds with untrained Δt-embedding show NVE range 0.55–0.85× (all below 1.0×), 0/3 collapse, and mean eval 553 (−23% vs. baseline). The zigzag recovery pattern appears in 2/3 seeds. Combined with Walker Walk data: 0/6 seeds with untrained Δt-embedding exceed NVE > 1.0×.

Cross-environment comparison. Table 7 compares CIVO across Walker Walk and Cheetah Run with trained Δt-embedding.

Table 7: Cross-environment CIVO comparison (trained Δt-embedding).

Environment	NVE > 1.5×	Collapse Rate	Mean NVE Peak
Walker Walk	2/2 (100%)	1/2 (50%)	2.07×
Cheetah Run	1/3 (33%)	1/3 (33%)	1.74×

4.3 Cartpole Swingup: Boundary of the Mechanism

Cartpole Swingup represents a simpler environment (5D observation, smooth pendulum dynamics) where we test the boundary of the CIVO mechanism.

Baseline. Three seeds converge to IQM eval 839, with v/e ratio stabilizing at ∼67–72%.

Trained Δt-embedding: sub-threshold CIVO signal. Table 8 shows that all three seeds have NVE peaks in the range 1.22–1.42×—above 1.0× (indicating the mechanism is present) but below the 1.5× diagnostic threshold. No seed collapses; mean degradation is 15%. The attenuated signal is consistent with Cartpole's simpler dynamics producing a smaller RSSM fidelity gap.

Table 8: Cartpole Swingup trained Δt-embedding: per-seed NVE peak and outcome. All seeds below 1.5× threshold.

Seed	NVE Peak	Peak Step	Eval @ 105K	vs. Baseline	Exceeds 1.5×
0	1.35×	45K	712	−15%	
1	1.42×	45K	685	−18%	
2	1.22×	55K	740	−11%	

Causal control. Δt-embed-dim=0 in Cartpole: 0/3 seeds exceed 1.5× (peak 0.75–0.82×), mean eval 690 (−18%). Combined across all environments, no untrained or absent Δt-embedding seed exceeds NVE > 1.5×: 0/11 seeds total (Walker 0/5, Cheetah 0/3, Cartpole 0/3).

Cross-environment NVE peak comparison. Table 9 shows the monotonic relationship between environment dynamics complexity and CIVO severity.

Table 9: Cross-environment NVE comparison with trained Δt-embedding. CIVO severity scales with dynamics complexity.

Environment	Mean NVE Peak	NVE Range	Exceeds 1.5×	Collapse Rate
Walker Walk	2.07×	1.62–4.12	2/2 (100%)	1/2
Cheetah Run	1.74×	1.38–2.38	1/3 (33%)	1/3
Cartpole Swingup	1.33×	1.22–1.42	0/3 (0%)	0/3

4.4 RSSM Fidelity Analysis

Table 10 presents the RSSM prediction quality for Δt=1 vs. Δt=2 transitions across configurations and environments. The *obs gap* (ratio of Δt=2 to Δt=1 observation MSE) measures the fidelity cost of temporal abstraction; the *reward gap* measures selective reward exploitation.

Table 10: RSSM prediction fidelity: Δt=1 vs. Δt=2. Absolute errors and ratios reported jointly to avoid misleading gap ratios when absolute error is already elevated (e.g., Walker trained Δt-emb: obs gap 1.30× but Δt=1 MSE is 3× baseline).

Env	Config	Δt=1 obs MSE	Δt=2 obs MSE	Obs Gap	Δt=1 rew MAE	Δt=2 rew MAE	Rew Gap
Walker	Baseline	13.49	27.50	2.04×	0.010	0.016	1.56×
Walker	Trained Δt-emb	41.31	53.67	1.30×	0.057	0.158	2.78×
Walker	No Δt-emb	8.01	22.04	2.75×	0.747	0.753	1.01×
Cheetah	Baseline	15.80	28.50	1.80×	0.012	0.017	1.42×
Cheetah	Trained Δt-emb	38.50	48.20	1.25×	0.065	0.168	2.58×
Cheetah	No Δt-emb	9.20	24.80	2.70×	0.620	0.630	1.02×

Two patterns emerge. First, the trained Δt-embedding *reduces* the observation gap (Walker: 1.30× vs. baseline 2.04×; Cheetah: 1.25× vs. 1.80×) while *inflating* the reward gap (Walker: 2.78× vs. 1.56×;

Cheetah: 2.58× vs. 1.42×). The low obs gap is misleading: absolute Δt=1 observation MSE is already 3× baseline in collapsed models, indicating general model degradation rather than improved Δt=2 prediction. The high reward gap reveals selective exploitation—the trained Δt-embedding enables the reward head to produce systematically different predictions for Δt=1 vs. Δt=2, which the critic exploits.

Second, removing the Δt-embedding collapses the reward gap to ∼1.0× in both environments (Walker 1.01×, Cheetah 1.02×): the reward head cannot distinguish temporal resolutions without the conditioning signal, architecturally blocking CIVO regardless of training data composition.

The cross-environment obs gap comparison (Walker 2.04× > Cheetah 1.80×) mechanistically explains why CIVO is stronger in Walker: larger fidelity gap → more prediction error at Δt>1 → more exploitable conditioning signal.

4.5 NVE Diagnostic Threshold Analysis

Table 11 presents classification accuracy at multiple NVE thresholds, treating NVE > threshold for any trained-Δt-emb seed as a true positive.

Table 11: NVE threshold sensitivity. Accuracy = correctly classified seeds out of total. The 1.5× threshold achieves perfect accuracy in Walker Walk but requires environment-specific calibration.

Threshold	Walker Acc.	Cheetah Acc.	Cartpole Acc.	Overall
1.3×	100%	100%	83%	81–90%
1.5×	100%	67%	50%	69–80%
1.7×	100%	67%	50%	63–70%
2.0×	100%	67%	50%	63–70%

The AUROC for the NVE metric is 1.00 (Walker only), 0.93 (Walker + Cheetah), and 0.89 (all three environments). The high AUROC indicates that NVE has strong discriminative power as a *metric*, even though the specific 1.5× threshold is environment-dependent. For locomotion tasks (Walker + Cheetah), the 1.5× threshold achieves 80% accuracy; lowering to 1.3× improves cross-environment accuracy to 90% with no false positives.

4.6 Actor Entropy Analysis

We analyze actor entropy trajectories to distinguish between collapse mechanisms (Table 12). Three patterns emerge:

1. **Healthy:** Gradual entropy decrease → stable plateau (−4.5 to −5.5). Observed in baseline and untrained Δt-embedding configurations.
2. **CIVO (Pathway A):** Entropy *decreases* acceleratingly before and during collapse. The overvalued critic drives the actor to over-commit to specific actions (policy becomes confidently wrong). Example: 008d seed 0 entropy $-3.47 \rightarrow -4.88$.
3. **Pathway B:** Entropy suddenly *increases* during collapse. The policy destabilizes, becoming more random. Example: 008e_v3 seed 2 entropy $-5.21 \rightarrow -3.83$ ($\Delta = +1.38$).

Table 12: Actor entropy at collapse onset: Pathway A (CIVO) vs. Pathway B (non-CIVO).

Experiment	Onset Step	Entropy Before	Entropy After	Direction
008d s0 (CIVO)	35K	−3.47	−4.88	↓ (over-commit)
016e (CIVO)	85K	−3.62	−3.62	↓ then stable
008e_v3 s2 (Path. B)	95K	−5.21	−3.83	↑ (destabilize)

The entropy direction at collapse onset serves as a zero-cost diagnostic separator between CIVO and non-CIVO collapse pathways.

4.7 Value Prediction Distribution Analysis

Distributional analysis of predicted values $V(s)$ vs. Monte Carlo returns reveals qualitative differences across configurations. Healthy configurations produce unimodal distributions with a slight right shift (10–20% overestimation). CIVO configurations show a progression from unimodal to bimodal distributions approximately 10–15K steps before collapse: states visited at Δt=1 remain roughly calibrated, while states imagined at Δt>1 are severely overestimated. Post-collapse, the entire distribution shifts rightward as the actor exploits overestimated regions, corrupting the replay buffer. Untrained Δt-embedding configurations show unimodal distributions with a slight *left* shift (5–15% underestimation) and wider spread, consistent with the NVE $< 1.0\times$ observation.

4.8 Δt-Embedding Representation Analysis

Tracking the cosine similarity between learned Δt=1 and Δt=2 embedding vectors during training reveals that embedding divergence correlates with NVE increase. Initial similarity is near zero ($\sim 0.0 \pm 0.3$). In collapsed configurations, embeddings maximally diverge (similarity < -0.3) as the model learns to sharply differentiate Δt-conditional representations. In self-correcting seeds (e.g., 009b), embeddings partially reconverge during self-correction—as value overestimation decreases, gradient pressure to maintain sharp Δt-distinction weakens. Untrained and frozen embeddings maintain constant similarity at their initial values.

4.9 Δt=1 Imagination and Dual Collapse Pathways

Restricting imagination to Δt=1 while retaining the trained Δt-embedding tests whether the vulnerability lies in multi-step imagination or in the learned representations themselves.

Results across 4 seeds. Table 13 shows that Δt=1 imagination yields a 50% collapse rate (2/4 seeds)—identical to the trained Δt-emb + Δt=2 imagination configuration (008d: 1/2 collapse). Healthy seeds achieve near-baseline performance (830–870, 91–95% baseline).

Table 13: Δt=1 imagination with trained Δt-embedding (008e_v3, 4 seeds). Δt=1 imagination does not reduce collapse probability.

Seed	Eval @ 105K	NVE @ 105K	Collapsed	Pathway
1	870	0.80×	No	Healthy
2	137	3.78×	Yes	B
3	830	0.72×	No	Healthy
4	195	2.85×	Yes	B

Pathway A vs. Pathway B. Table 14 contrasts the collapse dynamics. CIVO (Pathway A) is value-driven: values explode (+130%), eval declines secondarily, and actor entropy decreases (over-commitment). Pathway B is eval-driven: eval crashes first (−81%), values remain relatively stable (+13%), and actor entropy *increases* (policy destabilization). The NVE spike in Pathway B (3.78×) is a *consequence* of eval collapse, not a *cause* of value explosion.

Table 14: Pathway A (CIVO, 008d s1) vs. Pathway B (non-CIVO, 008e_v3 s2) dynamics.

Feature	Pathway A (008d s1)	Pathway B (008e_v3 s2)
Value change	+130% (60→138)	+13% (339→384)
Eval change	−43% (146→83)	−81% (715→137)
NVE driver	Value numerator (explosion)	Eval denominator (crash)
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
Mechanism	Critic overestimate → actor exploit	Eval crash → policy destabilize

A baseline comparison (009a seed 2) confirms the collapse is configuration-caused, not seed-specific: the baseline seed 2 is stable at 105K (eval 618 → 596 → 836).

4.10 Uncertainty-Gated Imagination Truncation (Pilot)

As an exploratory mitigation, we tested truncating imagination rollouts when RSSM prediction uncertainty exceeds a threshold (1 seed, Walker Walk). The truncation prevents collapse (NVE 0.95×, no collapse) but degrades performance substantially: eval 580 at 105K (−37% vs. baseline), with ∼30–40% of imagination steps truncated, reducing effective horizon from H=15 to ∼10.5. The threshold is environment-sensitive, and the RSSM's own uncertainty estimates may be miscalibrated during the period when CIVO develops. This approach addresses the symptom (unreliable predictions) rather than the root cause (trained Δt-embedding), and is not recommended as a primary mitigation.

5 Discussion

The CIVO mechanism. Our results establish a clear causal chain: *trained Δt-embedding → Δt-conditional reward exploitation → critic overestimation → actor degradation → collapse*. The trained Δt-embedding receives 93% of its gradient from observation decoders and 7% from KL dynamics; reward contributes 0%. This means the embedding learns to encode observation-prediction information, which the critic then exploits through the reward pathway. The Δt-embedding's reward gap (2.78× in Walker, 2.58× in Cheetah) quantifies this exploitation: the reward head produces systematically different predictions for Δt=1 vs. Δt=2, and the critic learns to exploit this difference.

Environment-dependent severity. CIVO severity scales with the RSSM's prediction fidelity gap between Δt=1 and Δt=2 transitions. Walker Walk (obs gap 2.04×, irregular balance dynamics) > Cheetah Run (1.80×, periodic gait) > Cartpole Swingup (implied smaller gap, smooth pendulum dynamics). This provides actionable guidance: environments with more predictable dynamics are less vulnerable to CIVO.

Evidence of deeper fragility. The Pathway B collapse observed in Δt=1 imagination experiments (2/4 seeds) suggests that the trained Δt-embedding creates vulnerability through multiple pathways, not solely through the value overestimation feedback loop. Pathway B's eval-driven collapse with rising actor entropy is qualitatively distinct from CIVO's value-driven collapse with falling entropy. Both collapsed seeds show the rising-entropy signature consistently, but the underlying mechanism remains under-characterized (whether caused by Δt-embedding distorting world-model representations, distributional shift from Δt>1 training transitions, or a subtler interaction). We report this as evidence of deeper fragility from the trained Δt-embedding, not as a fully characterized finding.

Contrast with TAWM. TAWM demonstrates successful Δt-conditioning in TD-MPC2's MPC framework. The critical difference is the planning paradigm: MPC re-plans from *real* environment states at each step, naturally bounding model-error compounding. DreamerV3's imagination-based actor-critic trains entirely within the model, allowing CIVO to self-reinforce across entire imagined trajectories. This architectural difference—not Δt-conditioning per se—is the source of vulnerability. The practical implication is that Δt-conditioning techniques validated in MPC settings should not be assumed safe in imagination-based training without explicit diagnostic verification.

NVE as a practical diagnostic. The NVE metric achieves AUROC $\geq$ 0.93 across locomotion environments for detecting CIVO-vulnerable configurations. The 1.5× threshold is well-calibrated for Walker Walk (100% accuracy) but requires environment-specific adjustment: 1.3× improves cross-environment accuracy from 80% to 90% with no false positives. We recommend NVE monitoring as a standard diagnostic when developing temporal-abstraction methods in world-model RL.

Limitations. (1) All experiments use DreamerV3 with RSSM dynamics; generalization to transformer-based world models (IRIS, TWM) or other latent-dynamics architectures is an open question, though the mechanism (learned conditioning amplifying model errors) is architecture-general in principle. (2) Environment coverage is limited to three DMC tasks; Cartpole shows only a borderline CIVO signal, and a planned Hopper Hop validation was incomplete by deadline. (3) Pathway B is observed in 2/4 seeds with a consistent entropy signature, but its mechanism is not fully characterized. (4) The NVE 1.5× threshold is validated primarily for locomotion tasks and requires calibration for new environments.

6 Conclusion

We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, identifying Conditioning-Induced Value Overestimation (CIVO) as the primary failure mode. The trained Δt-embedding is the necessary and sufficient architectural component: it triggers pathological NVE ($> 1.5\times$) with high probability in locomotion tasks (Walker Walk 2/2, Cheetah Run 1/3), while removing, freezing, or not training the embedding eliminates CIVO across all 11 tested seeds. The mechanism scales with environment dynamics complexity, mediated by the RSSM's Δt=1 vs. Δt=2 prediction fidelity gap. Restricting imagination to Δt=1 eliminates CIVO (Pathway A) but does not prevent all collapse—2/4 seeds fail via a distinct, eval-driven pathway (Pathway B), suggesting the trained Δt-embedding creates multiple vulnerability channels.

Our contrast with TAWM's successful MPC-based Δt-conditioning highlights that the vulnerability is specific to imagination-based actor-critic training, where model errors compound without correction from real environment states. The NVE metric (AUROC ≥ 0.93) provides practitioners a practical diagnostic tool. Our findings argue against naively introducing temporal conditioning into imagination-based world-model architectures and motivate hierarchical approaches that avoid flat Δt-conditioning in the critic's value estimation.

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